

TABLE 2
EXTRACT FROM SK6018CD4-28 DATASHEET

Parameter	Symbol	Test Conditions	Min	Typ	Max	Unit
EN Input Current	I _{EN}	V _{EN} = 5.5V	--	--	0.1	μA
Line Regulation	ΔLINE	I _{LOAD} = 1mA	1.2V ≤ V _{IN} < 1.5V	--	0.3	%
			1.5V ≤ V _{IN} < 1.8V	--	0.15	
			1.8V ≤ V _{IN} < 5.5V	--	0.13	
Load Regulation	ΔLOAD	1mA < I _{LOAD} < 600mA	--	0.5	1	%
Power Supply Rejection Ratio	PSRR	V _{IN} = 3V, I _{LOAD} = 50mA, C _{OUT} = 1μF, V _{OUT} = 2.5V, f = 1kHz	--	75	--	dB
Output Voltage Noise		C _{OUT} = 1μF, I _{LOAD} = 150mA, BW = 10Hz to 100kHz, V _{IN} = V _{OUT} + 1V	V _{OUT} = 0.8V	--	26	μVRMS
			V _{OUT} = 1.2V	--	37	
			V _{OUT} = 1.8V	--	39	
			V _{OUT} = 3.3V	--	42	
Output Current Limit	I _{LIM}	V _{OUT} = 90%V _{OUT(Normal)}	610	1100	--	mA
Enable Input Voltage	Logic-High	V _{IH}	V _{IN} = 5V	0.9	--	V
	Logic-Low	I _L	V _{IN} = 5V	--	0.4	
Thermal Shutdown Temperature	T _{SD}	I _{LOAD} = 30mA, V _{IN} ≥ 1.5V	--	150	--	°C
Thermal Shutdown Hysteresis	ΔT _{SD}		--	20	--	°C
Discharge Resistance		EN = 0V, V _{OUT} = 0.1V	--	80	--	Ω

(Courtesy: SK6018CD4-28 datasheet)

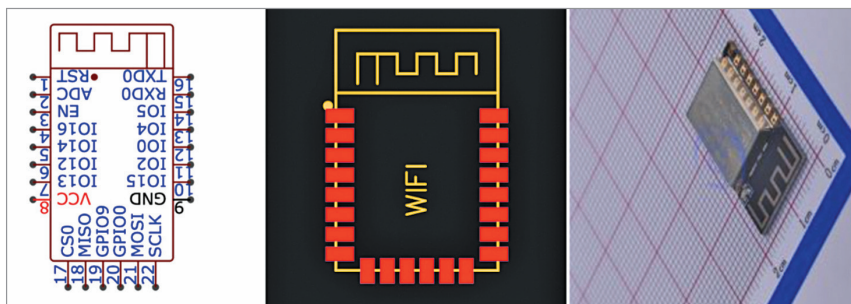


Fig. 4: The ESP12F chip footprint and pinout

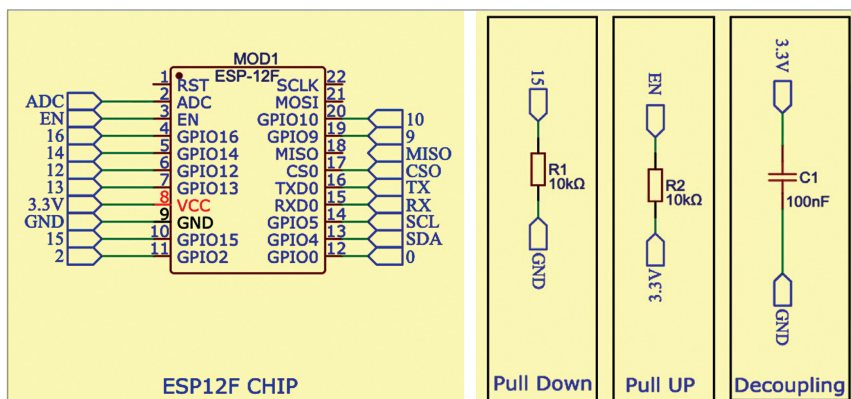


Fig. 5: ESP12F chip design configuration

The design also offers flexibility in display options. Users can select an OLED display (over I²C) or a colour display such as the ST7789V round-edge, GC9A01, or any other compatible module. Both types can operate simultaneously, with OLED on I²C and colour displays on SPI.

Power regulation is managed by an LDO regulator, ensuring stable operation across all components.

All components are SMD to maintain a small, compact form factor, with only the buttons using through-hole footprints. Fig. 1 shows the author's prototype, while

Fig. 2 compares the market-available Game Boy with the author's design. Required components are listed in the Bill of Materials (Table 1).

Design

The primary requirement of the device is a power supply capable of operating the entire system.

The ESP12F supports an operating voltage range of 3.0V to 3.3V, and the display operates within the same range. Therefore, a 3.3V regulated supply is required to provide a stable voltage, regardless of whether the input source is a 3.7V Li-ion/LiPo battery, a 5V or 7V battery, or USB 5V power. The regulator ensures a clean and stable 3.3V output for the entire system.

An LDO regulator is employed for this purpose. While a common AMS1117-3.3 or similar 3.3V LDO could be used, the SK6018CD4-28 was selected to maintain a compact form factor. It comes in a very small DFN-4 (1mm × 1mm) package, saving PCB space. The datasheet specifies an input voltage range of 1.2V to 5.5V, delivering an output current of up to 600mA (V_{IN} ≥ 2.3V with EN enabled). This is sufficient to power the ESP12F during peak processing and Wi-Fi transmission, as well as the display module. Table 2 shows an extract from the SK6018CD4-28 datasheet.

The LDO requires two 1μF ceramic capacitors (one between V_{IN} and GND, and one between V_{OUT} and GND) to ensure stability and proper filtering. SMD ceramic capacitors are used to maintain compactness. Refer to Fig. 3 for the LDO voltage regulator circuit.

Game Boy core ESP12F design

According to the datasheet, the ESP12F provides 22 GPIO pins, including three strapping pins (GPIO0, GPIO2, and GPIO15) that determine the boot configuration. For normal boot mode, GPIO15 must be pulled to GND via a resistor. During programming or firmware upload,